

Wf: 6.8

$$v = E(T(X)) ; X = X_1, \dots, X_n \text{ i.i.d.} ; X \sim U[3, v]$$

$$X_i \sim U[3, v] \Rightarrow \bar{X} \sim U[3, v] \Rightarrow E(\bar{X}) = \frac{3+v}{2} \Leftrightarrow v = 2E(\bar{X}) - 3$$

$$E(T(X)) = 2E(\bar{X}) - 3 = E(2\bar{X} - 3) \Rightarrow T(X) = 2\bar{X} - 3$$

$$\hat{v} = 2\bar{X} - 3 \quad \hat{v} = \frac{2}{n} \sum_{i=1}^n X_i - 3$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Wf: 7.2/a) b) c) ; 7.6/d)

7.2 a) $X_i \sim N(\mu, \sigma^2) \quad \sigma \in \mathbb{R}$

$$\begin{aligned} \underline{ML}: L(x|\mu) &= \prod_{i=1}^n f_{\mu}(x_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) = \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \prod_{i=1}^n \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right) \end{aligned}$$

$$\ell(x|\mu) = \ell(\mu) = \ln L(\mu) = n \ln\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

$$\ell'(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0$$

$$\sum_{i=1}^n (x_i - \mu) = 0$$

$$-n\mu + \sum_{i=1}^n x_i = 0$$

$$\sum_{i=1}^n x_i = n\mu \Leftrightarrow \frac{1}{n} \sum_{i=1}^n x_i = \mu = \bar{x}$$

$$\underline{ML}: \hat{\mu} = \bar{x}$$

$$\underline{MM}: \frac{1}{n} \sum_{i=1}^n x_i = m_1(x) = M_1(X, \mu) = E(X) = \mu \Rightarrow \underline{MM}: \hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

b) $X_i \sim \text{Bin}(m, p) \quad m \in \mathbb{N}$

$$\underline{ML}: L(x|p) = \prod_{i=1}^n P(X=x_i) = \prod_{i=1}^n \binom{m}{x_i} \cdot p^{x_i} (1-p)^{m-x_i} = \prod_{i=1}^n \binom{m}{x_i} p^{\sum_{i=1}^n x_i} (1-p)^{nm - \sum_{i=1}^n x_i}$$

$$\ell(x|p) = \ell(p) = \ln L(p) = \ln\left(\prod_{i=1}^n \binom{m}{x_i}\right) + \left(\sum_{i=1}^n x_i\right) \ln(p) + \left(nm - \sum_{i=1}^n x_i\right) \ln(1-p)$$

$$\ell'(p) = \frac{\sum_{i=1}^n x_i}{p} - \frac{nm - \sum_{i=1}^n x_i}{1-p}$$

$$\frac{\sum_{i=1}^n x_i}{p} - \frac{nm - \sum_{i=1}^n x_i}{1-p} = 0 \Leftrightarrow \frac{1-p}{p} = \frac{nm - \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i} \Leftrightarrow$$

$$\underline{ML}: \hat{p} = \frac{\bar{x}}{m}$$

$$\Leftrightarrow \frac{1}{p} - 1 = \frac{nm}{\sum_{i=1}^n x_i} - 1 \Leftrightarrow p = \frac{\bar{x}}{m}$$

$$\underline{MM}: \bar{x} = m_1(x) = M_1(x, p) = E(X) = mp \Leftrightarrow p = \frac{\bar{x}}{m} \Rightarrow \underline{\underline{MM: \hat{p} = \frac{\bar{x}}{m}}}$$

$$c) X_i \sim \text{Exp}(\lambda)$$

$$\underline{ML}: L(x, \lambda) = \prod_{i=1}^n f_{\lambda}(x_i) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n \cdot e^{-\lambda \sum_{i=1}^n x_i}$$

$$\ell(x, \lambda) = \ell(\lambda) = \ln L(\lambda) = n \cdot \ln(\lambda) - \lambda \cdot \sum_{i=1}^n x_i$$

$$\ell'(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n x_i \quad \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0 \Leftrightarrow \lambda = \frac{1}{\bar{x}}$$

$$\underline{ML: \hat{\lambda} = \frac{1}{\bar{x}}}$$

$$\underline{MM}: \bar{x} = m_1(x) = M_1(x, \lambda) = E(X) = \frac{1}{\lambda} \Rightarrow \underline{\underline{MM: \lambda = \frac{1}{\bar{x}}}}$$

7.6

$$d) X_i \sim N(\mu, \eta) \quad \eta = \sigma^2$$

$$2 u_{1-\alpha} \cdot \frac{\sigma}{\sqrt{n}} < 1 \Leftrightarrow 4 u_{1-\alpha}^2 \cdot \sigma^2 < n \Leftrightarrow 43,2887 < n$$

$$x_1 = 15,57$$

$$x_2 = 10,88$$

$$x_3 = 9,68$$

$$x_4 = 11,55$$

$$x_5 = 11,43$$

$$\alpha = 0,05 \Rightarrow u_{0,95} \approx 1,645 \quad R\text{-ben } \text{qnorm}(0,95)$$

legálabb 44 elemű minta kell